STAFF REVIEW · H446 033 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. False. AND (&&) is true only when both conditions are true.

2. Q remains 1. The input is sampled only when the triggering clock edge occurs.

3. B, C, D. A leaves first because a queue is FIFO.

4. 7 hashes to 2 and is stored at 2. 12 also hashes to 2, so probes to 3. 17 also hashes to 2; 2 and 3 are occupied, so it goes to 4.

I Do — source worked steps

1. Create the buckets — CODE:
```
const int SIZE = 7;
string?[] usernames = new string?[SIZE];
```

2. Write the index function — CODE:
```
static int Hash(string key, int size)
{
int total = 0;
foreach (char letter in key)
total += (int)letter;
return total % size;
}
```

3. Explain the limitation — CODE:
```
// Hash("ab", 7) == Hash("ba", 7)
// Same total -> collision.
// This is an educational hash function,
// NOT a secure password hash.
```

We Do — guided CHECK

1. Insert with linear probing
CODE:
```
int index = Hash(username, SIZE);
while (usernames[index] != null)
{
index = (index + 1) % SIZE;
}
usernames[index] = username;
```

2. Explain your solution first
How can we find a name after a collision?
Search using the same probing sequence

CODE:
int index = Hash(username, SIZE);
int checkedBuckets = 0;
while (checkedBuckets < SIZE && usernames[index] != null)
{
if (usernames[index] == username) break;
index = (index + 1) % SIZE;
checkedBuckets++;
}

3. Explain your solution first
With table size 7, why do 'ab' and 'ba' choose the same initial index?
Predict a collision
Both strings contain the same character codes, so the total is identical. Linear probing moves the second username to the next free bucket.

You Do — source mark schemes

1. CODING 1 · INSERT
string?[] usernames = new string?[7]; static int Hash(string key, int size) { int total = 0; foreach (char c in key) total += c; return total % size; }. Insert only if the username is not already present and a bucket is available.

2. CODING 2 · COLLISION
The ASCII/Unicode scalar totals for 'ab' and 'ba' are equal, so both yield the same index. For table size 7, (97 + 98) % 7 = 6. If 'ab' uses bucket 6, 'ba' wraps to bucket 0 using (6 + 1) % 7.

3. CODING 3 · LOOKUP
Recalculate the starting hash and probe, stopping when a matching username is found, a null bucket is reached, or seven positions have been checked. A full cycle must terminate. Never assume each username lives at the original bucket.

4. SYNOPTIC MCQ BANK
Question-bank checklist: linked-list follows pointers; directed graph edges have direction; stack LIFO; queue FIFO; tree is acyclic hierarchy; BST orders smaller left and larger right; hash table uses keys, hashing and collision resolution. Explanations are required.

Exit Ticket — source answer

1. False. A simple character-code sum is not a secure password hash. It is only suitable for demonstrating bucket lookup.